Project 3

An AI-Powered Diabetes Care and Follow-Up Assistant

Adeed | عضيد



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Executive Summary

Diabetes management involves more than collecting glucose readings. Patients may need to understand technical report metrics, compare results across time, organize relevant laboratory information, and prepare questions for clinical appointments. A digital service that brings these tasks into one structured workflow may help patients engage with their information more confidently. However, the extent of this need among the intended users must be validated rather than assumed.

Adeed is a proposed AI-assisted diabetes-care and follow-up service designed to fit within a Sehhaty-inspired digital health experience. The concept combines supported glucose-report extraction, data validation, comparison with compatible historical results, conversational explanations, appointment preparation, and a structured pathway for professional review.

The International Diabetes Federation estimates that 5.3 million adults aged 20-79 were living with diabetes in Saudi Arabia in 2024, representing an age-standardised prevalence of 23.1% (International Diabetes Federation [IDF], 2025). This establishes the public-health relevance of exploring practical digital support, but it does not by itself prove the specific usability problem addressed by Adeed.

The proposed design deliberately separates deterministic processing from generative AI. Numerical calculations, unit checks, historical comparability, and workflow state changes should be handled by tested application logic. AI should be limited to explaining verified information, answering general educational questions, and helping patients prepare questions for a clinician. It must not diagnose, prescribe insulin doses, or independently change treatment.

Adeed is a concept and prototype proposal, not an officially endorsed Ministry of Health or Sehhaty feature. Its value should be assessed through problem validation, extraction and comparison testing, usability studies, safety evaluation, and feedback from patients and healthcare professionals.

1. Target Organization

1.1 Organization Overview

The Saudi Ministry of Health (MOH) provides healthcare services and supports digital health services in the Kingdom. The Ministry describes Sehhaty as a unified platform through which users can access health information and services, including appointment-related services, medical reports, remote consultations, and selected health and wellness services (MOH, n.d.).

1.2 Rationale for Selection

Diabetes care can involve repeated glucose monitoring, laboratory testing, and discussions with healthcare professionals. A service that organizes selected results and helps patients prepare for follow-up is therefore a relevant design opportunity. The IDF's 2025 Diabetes Atlas reports an estimated 5.3 million adults with diabetes in Saudi Arabia in 2024, highlighting the scale of the condition (IDF, 2025). The proposed project focuses on one defined workflow rather than redesigning the entire platform.

1.3 Existing Service Opportunity

The opportunity explored by this project is a connected workflow for reviewing glucose reports, comparing compatible historical measurements, preparing questions, and navigating professional follow-up. Since Sehhaty already provides several digital health services, the project must not claim that appointment booking, medical reports, or consultations are absent. Instead, the specific gap - whether patients need a more consolidated glucose-report interpretation and preparation workflow - should be tested through official service review and user research.

2. Industry Background

Diabetes self-management may involve information from continuous glucose monitoring (CGM), insulin-pump or sensor reports, laboratory tests such as HbA1c, and previous care recommendations. These data sources are not interchangeable: they may cover different time periods, use different units, and measure different aspects of glycaemia.

The American Diabetes Association's Standards of Care discuss CGM metrics such as time in range, time above range, and time below range as useful elements of glycaemic assessment. They also describe the importance of sufficient CGM data when interpreting these measures (American Diabetes Association Professional Practice Committee, 2025). Adeed should therefore display each metric with its date range, unit, source, and any relevant data-quality limitation.

AI may support document understanding and plain-language explanations, but healthcare applications require safeguards, accountability, transparency, and human oversight. The World Health Organization's guidance on ethics and governance of AI for health emphasizes that ethical principles and human rights should be central to the design and use of AI systems (World Health Organization [WHO], 2021). Accordingly, Adeed is designed as an informational and coordination aid - not an autonomous clinical decision-maker.

3. Problem Validation

3.1 Problem Context

People managing diabetes may need to interpret glucose sensor or insulin-pump reports, understand changes in their readings over time, and prepare questions for clinical consultations. These tasks can involve several metrics, different reporting periods, and laboratory measurements that should not always be interpreted in the same way.

The proposed Adeed feature addresses this potential challenge by organizing supported glucose-report metrics, comparing compatible historical results, and helping patients prepare for conversations with healthcare professionals.

However, the relevance of diabetes as a public-health issue does not, by itself, establish that patients experience this specific difficulty. The proposed user problem must therefore be examined using credible research and, where feasible, direct feedback from intended users.

3.2 Evidence and Its Limits

The following sources provide context for the project and help establish why the proposed problem is worth investigating. Each item states what it shows and what it does not.

Evidence 1: Diabetes burden in Saudi Arabia

The International Diabetes Federation's Diabetes Atlas provides country-level estimates of diabetes prevalence and population burden. This information establishes the public-health relevance of exploring services that support people living with diabetes.

Relevance to Adeed: The scale of diabetes in Saudi Arabia provides a rationale for investigating whether a focused digital service could help patients understand and organize their glucose-report information.

Limitation: Prevalence statistics do not demonstrate that patients struggle to interpret reports, compare historical results, or prepare questions for appointments.

Evidence 2: Complexity of glucose-monitoring metrics

The American Diabetes Association's Standards of Care discuss the use of continuous glucose monitoring metrics, including time in range, time above range, and time below range, in diabetes assessment.

Relevance to Adeed: These metrics provide a clinical context for the proposed feature's plain-language explanations and structured report presentation.

Limitation: Clinical guidance establishes the relevance of these measurements, but it does not prove that patients find them difficult to understand.

Evidence 3: Digital healthcare and AI governance

The Saudi Ministry of Health provides digital healthcare services through Sehhaty. Separately, the World Health Organization's guidance on AI for health emphasizes the importance of responsible governance, transparency, and human oversight.

Relevance to Adeed: These sources inform the proposed digital-service context and support the need to design AI assistance with appropriate safeguards.

Limitation: Neither source establishes that the exact Adeed workflow is missing from current services or that users would adopt it.

Evidence 4: Saudi study on glucose-monitoring knowledge among CGM users

A cross-sectional questionnaire study of Saudi adults with type 2 diabetes who use continuous glucose monitoring devices (Jehad et al., 2026, Bahrain Medical Bulletin) reported mixed views of monitoring devices: 31.5% found them convenient, 31.1% were indifferent, and 37.7% disagreed with their value. Only 59.4% of participants consistently monitored their blood sugar during hypoglycaemia. The authors identified "a significant knowledge gap" among Saudi patients with type 2 diabetes regarding hypoglycaemia awareness and management, and concluded that educational and technological solutions are crucial.

Relevance to Adeed: It is a Saudi-specific signal that people who already use glucose-monitoring technology may still need educational support, which is the role of Adeed's plain-language explanations.

Limitation: The study measured hypoglycaemia awareness and device satisfaction, not the ability to interpret CGM or pump reports. Only the published abstract was reviewed, and the sample size is not stated there. It supports the case for patient education, not the specific Adeed workflow.

Evidence 5: Patient-reported confusion with the AGP report (community forum)

In a public thread titled "Ambulatory Glucose Profile" on the diabetes.co.uk community forum (13 September 2021), a forum member wrote: "I'm confused because the numbers on the left don't seem to match the graphics on the right." The remark concerns the layout of the standard AGP report, which places percentage figures beside a colour bar.

Relevance to Adeed: It is a real example of a person living with diabetes struggling to read the very report format that Adeed is designed to translate into plain language.

Limitation: It is a single anecdotal post from a UK-based forum, written in 2021 and in English. It does not show how common the difficulty is and says nothing about Saudi users. It is used only as an illustrative qualitative signal.

3.3 Evidence Gap and Further Validation

The available secondary evidence establishes the relevance of diabetes care and the importance of presenting health information responsibly. However, it does not yet establish the frequency or severity of the specific user problem addressed by Adeed.

Further validation should investigate how intended users currently access glucose reports, interpret their metrics, compare previous results, and prepare for appointments.

Social listening as supporting evidence

To complement formal research, the team will collect public posts in which people describe difficulty with glucose reports or appointment preparation, on platforms such as X, Reddit, and public Arabic-language diabetes communities. Posts are included only when they are public, dated, and linked, and are quoted verbatim without identifying the author. Social-media content is anecdotal and self-selected; it is used to generate hypotheses and illustrate themes, never to estimate prevalence. The log below records each item as it is verified.

#	Platform	Date	Language	Verbatim excerpt	Theme	Link
1	diabetes.co.uk forum	13 Sep 2021	English	"I'm confused because the numbers on the left don't seem to match the graphics on the right."	AGP layout is hard to read	diabetes.co.uk/forum/threads/ambulatory-glucose-profile.184023/page-2 (post #22)

If feasible, the team should also conduct a small exploratory study involving people who manage diabetes and healthcare professionals. Participants could be asked to complete a report-interpretation task using a de-identified sample report. The team could then document observed difficulties, questions raised, task completion, and participant feedback.

Any interviews or usability tests must be reported only after they have actually been conducted. The report should identify the research method, participant count, recruitment approach, findings, and limitations. A small convenience sample should not be presented as representative of the wider population.

Until this additional evidence is collected, the specific user problem should be treated as a research hypothesis rather than an established finding.

3.4 Problem Statement

People managing diabetes who use glucose sensor or insulin-pump reports may need a more convenient way to understand technical metrics, compare compatible historical results, and prepare focused questions for clinical consultations. Adeed proposes a guided digital workflow that brings these activities together and explains verified information in accessible language.

The project will evaluate whether this approach improves users' understanding and appointment preparation. The extent of the underlying need and the effectiveness of the proposed solution remain to be validated through user research and usability testing.

4. Competitive Benchmark

To position Adeed, we reviewed the public pages of established diabetes-data platforms and of the national platform it is designed to sit beside. The table records only what each page states; where a capability is not mentioned, the cell says "not stated", which does not prove the capability is absent. All pages were accessed on 10 November 2026.

Solution	Main audience	What the page says it does	Arabic	AI explanation of reports
Sehhaty (MOH)	All residents	Unified platform for health information and services, including appointment-related services, medical reports, and remote consultations (see 1.1).	Not reviewed	Not described in the cited MOH page; to be confirmed in the official service review.
Dexcom Clarity	Dexcom CGM users and clinics	Gives insights into retrospective glucose values, patterns, and trends; time-in-range goals; reports can be viewed, saved, printed, and emailed; clinics can be given access to data and reports.	Listed (the app lists Arabic among 28 languages)	Not stated on the listing.
Glooko	Patients, clinicians, care teams	Patient app syncs device data, logs context, shows trends, and shares with care teams. Clinicians can review data across 200+ compatible devices, with EHR documentation where integrated.	Not stated	Not stated.
Tidepool	Patients and clinicians (non-profit)	Combines data from many devices in one place, shares data with chosen people, offers insights, and has a hub to prepare for virtual visits.	Not stated	Not stated; "insights" are described generally.
mySugr	Patients	Logging of insulin, carbs, activity, and photos; estimated HbA1c; CSV reports (free) and PDF/Excel reports (PRO); sharing through the Accu-Chek Care Platform.	Not stated	Not stated.

4.1 Where Adeed Fits

These platforms are strong at collecting, storing, and sharing device data, and most are aimed at clinicians or built around one manufacturer's devices. Within the pages reviewed, none describes a single patient-facing flow that combines all of the following, which is the space Adeed explores:

- **Plain-language, Arabic-first explanation** of the metrics in a report the patient already holds, with the source page of every value.

- **Rule-based comparability checks** so that only compatible reports are compared, and limitations are shown instead of hidden.
- **GMI and HbA1c kept apart,** because they are different measures.
- **Appointment preparation** in the same flow: a one-page summary and ready-to-ask questions.
- **Bounded AI** that explains verified data, refuses dosing, and shows an emergency notice.
- **A structured, human-in-the-loop review pathway** (simulated in the prototype).

Adeed is complementary rather than competing: it reads PDF reports generated by existing devices and platforms, and does not replace them.

Note: *This comparison reflects the public pages reviewed and the prototype's design intent. It should be confirmed by hands-on testing of each product, and the Sehhaty row by the official service review described in section 1.3. LibreView was not included because its official page could not be verified at the time of writing.*

5. Target Users

5.1 Primary Users: People Living with Diabetes

The primary users are people who receive supported glucose sensor or insulin-pump reports and want help organizing information before a clinical appointment. They want to:

- Understand the meaning of supported report metrics in accessible language.
- Review current results alongside compatible previous reports and laboratory measurements.
- See when historical data is missing, incomplete, or unsuitable for comparison.
- Prepare questions for a healthcare professional.

5.2 Secondary Users: Healthcare Professionals

The secondary users are authorized healthcare professionals who may review patient-submitted reports and need a structured view of relevant results, historical changes, and data limitations. They want to:

- Review source information and data-quality warnings.
- Understand the reason for a patient's review request.
- Document recommendations or request additional information.
- Recommend appropriate follow-up or laboratory testing within their professional scope.

5.3 Other Stakeholders

Potential stakeholders include healthcare organizations, laboratory providers, digital health product teams, privacy and security teams, and administrators responsible for clinical governance. Their involvement would depend on the implementation and any future integrations.

6. Proposed AI Solution

Adeed is an AI-powered diabetes care and follow-up assistant designed to help people with diabetes understand their glucose reports, compare compatible historical results, and prepare for medical appointments.

The feature is designed to fit within the My Profile → Health Profile section of the existing Sehhaty-inspired application, as the fourth tile of the Health File grid (Figure 1). It combines automated report processing, structured data comparison, AI-assisted explanations, and appointment preparation in one guided user journey.

Adeed uses deterministic application logic to extract, validate, and compare numerical data, while AI supports plain-language explanations and question preparation. It is designed to support patients, not replace healthcare professionals.

How it works

1. The user opens Adeed from their health profile.
2. The user uploads a supported glucose-monitoring or insulin-pump PDF report.
3. The system extracts available metrics and validates the data.
4. The user reviews the report summary and compares compatible historical results.
5. The AI assistant, Sukkarah, explains verified information and helps prepare questions.
6. The user creates an appointment-preparation summary and follows the available review workflow.



Figure 1. *Entry point: the Adeed tile in the Health File section of My Profile.*

Note: *All screenshots show synthetic demonstration data for a fictional patient. They are not real patient records.*

7. Core Features

Feature 1: Glucose Report Extraction

Adeed processes supported PDF reports (PDF only, up to 10 MB; a report covering 14 days or more is preferred) and extracts available glucose-monitoring metrics. It displays values with their units, dates, and source information when available. The system distinguishes extracted, calculated, unavailable, and uncertain values to make data limitations visible.

User value: Reduces manual effort and organizes technical report information in one place.



Figure 2. *Reports list with the upload box and the report type recognised for each file.*

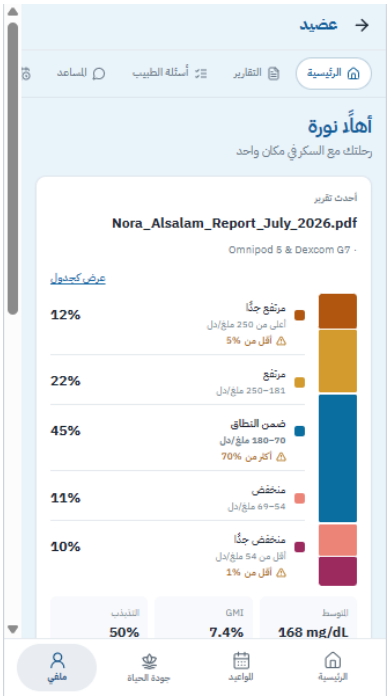


Figure 3. *Extracted metrics: time in each glucose range against the general targets.*

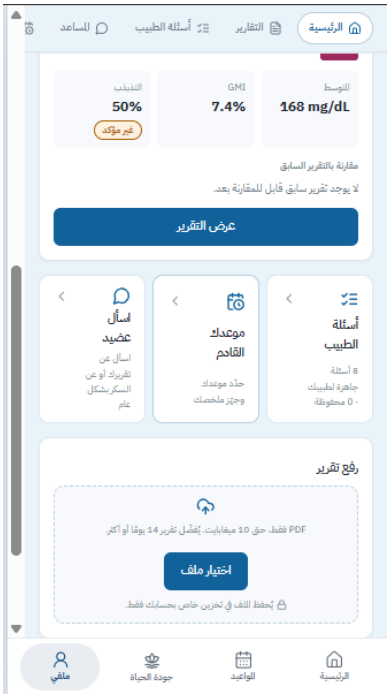


Figure 4. *Mean glucose, GMI, variability, and the comparison status.*

Feature 2: Historical Results Comparison

Adeed compares the current report with compatible previous reports and relevant laboratory results. It presents current and previous values, reporting periods, dates, units, and differences. When results are incomplete or unsuitable for comparison, the system clearly explains the limitation.

HbA1c laboratory results and Glucose Management Indicator (GMI) estimates are displayed separately because they are different measures.

User value: Helps users review changes over time and identify information they may want to discuss with their healthcare professional.

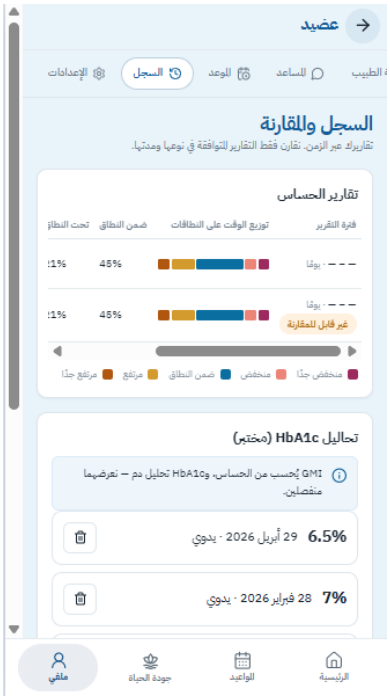


Figure 5. *History and comparison: sensor reports over time with a "not comparable" label, and HbA1c results shown separately.*

Feature 3: AI Patient Assistant - Sukkarah

Sukkarah provides plain-language explanations of verified report metrics, answers general diabetes-education questions, and helps users prepare questions for clinical consultations. Its responses should be grounded in available, verified information and acknowledge uncertainty or missing data.

The assistant does not diagnose conditions, prescribe insulin doses, or change treatment recommendations. In Figure 6 the user asks how much insulin to take for a meal; the assistant declines to give a dose and explains what to check and who decides.

User value: Makes technical information easier to understand and supports more informed conversations with healthcare professionals.



Figure 6. *Sukkarah declining to calculate an insulin dose and pointing to the care team.*

Feature 4: Appointment Preparation Summary

Adeed consolidates the current report summary, compatible historical comparisons, relevant available laboratory results, missing information, and questions the user wants to ask during an appointment.

The user can review the summary before a consultation. Real appointment booking is available only if a verified booking integration exists.

User value: Helps users organize relevant information and prepare focused questions before a medical appointment.



Figure 7. *Suggested questions for the doctor, each with a "why this question" explanation and the data it is based on.*



Figure 8. *Appointment preparation: date, clinic, personal notes, and a one-page summary.*

Feature 5: Review and Follow-Up Workflow

When supported by the application, the user can submit a review request and follow its status. An authorized healthcare professional can review submitted information, record a recommendation, or request additional details.

The interface must distinguish real review actions from simulated prototype states. In the prototype the review is simulated: a fixed, versioned demonstration rule set flags a report as needing specialist review, the request goes to a demonstration reviewer dashboard, and the decision is shown to the patient. No real clinician is involved.

User value: Provides a more structured way to organize report review and follow-up information.



Figure 9. *Patient view: a rule-based flag "needs specialist review" with the rules that triggered it.*



Figure 10. *Patient view: notifications showing the review request being sent and answered.*

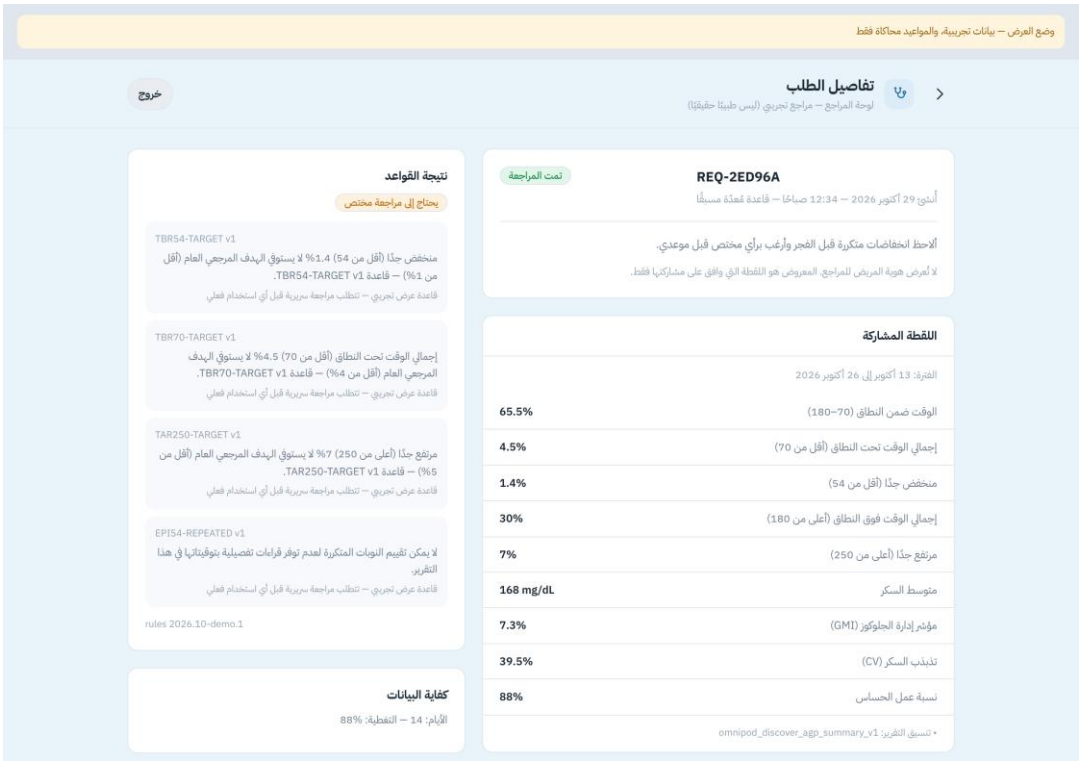


Figure 11. *Reviewer dashboard (simulated): request details, shared report snapshot, and rule results. Labelled "demo reviewer - not a real doctor".*

Note: *The review screens come from the earlier prototype build (28 October 2026) and the simulated reviewer dashboard, because the current web build does not include the review flow. Replace them with current screenshots once the flow is connected.*

8. User Journey

8.1 Current Journey (Research Hypothesis)

The following is a preliminary journey hypothesis and should be replaced or refined using evidence from user research or verified service observation.

1. The patient receives or downloads a glucose report.
2. The patient attempts to understand the report's technical metrics.
3. The patient locates previous glucose reports or laboratory results, if available.
4. The patient tries to identify meaningful changes across the records.
5. The patient prepares questions or concerns before a clinical appointment.
6. The patient discusses the information with a healthcare professional.

Potential friction points to investigate include unfamiliar terminology, manual comparison, missing historical data, uncertainty about whether measurements are comparable, and difficulty organizing questions. One of these (difficulty reading the report layout) has an illustrative example in Evidence 5; the others should not be reported as confirmed findings unless research supports them.

8.2 Proposed Adeed Journey (as implemented in the prototype)

The journey below follows the screens and routes of the working prototype. Figure numbers refer to the screenshots in sections 6 and 7.

Step	What the user does	What Adeed does	Screen
1. Enter	Opens My Profile and taps the Adeed tile in the Health File grid. First-time users create an account or sign in.	Shows a privacy statement: health data is private to the account and files are kept in private storage.	Figure 1
2. Upload	Chooses or drags in a glucose-report PDF.	Accepts PDF only, up to 10 MB, and suggests a report of 14 days or more. Shows upload and processing status.	Figures 2, 4
3. Review the summary	Reads the home summary of the latest report.	Shows time in each glucose range against general targets, mean glucose, GMI, and variability. Each value is marked extracted, calculated, unavailable, or uncertain.	Figures 3-4
4. Check the details	Opens the report to see where each number came from.	Shows the source page of each value, data quality, device settings (view only, never suggested for change), and the nearest HbA1c result.	Not yet captured
5. Compare	Opens History to see change over time.	Compares only compatible reports (same type and period) and labels limited or impossible comparisons. GMI and HbA1c are shown separately, and HbA1c results can be added manually.	Figure 5
6. Get the questions	Opens Doctor Questions.	Suggests questions based on the actual findings, each with a "why this question" explanation. The user can edit, save, add their own, and copy them.	Figure 7
7. Ask Sukkarah	Asks about the report or about diabetes in general.	Explains verified data in plain language. Refuses to diagnose or give doses, and shows an emergency notice when needed.	Figure 6
8. Prepare the visit	Enters the appointment date, clinic, and personal notes.	Builds a one-page summary with key findings, missing or uncertain data, HbA1c results, and the user's questions, ready to copy or print.	Figure 8
9. Request review (simulated)	Sends a review request when a report is flagged.	A demonstration rule set flags the report, a demo reviewer records a decision, and the patient is notified. No real clinician is involved.	Figures 9-11
10. Return	Comes back later with a new report.	Keeps the report history and shows the comparison against the previous compatible report.	Figure 5

8.3 Journey Comparison

The "today" column is the hypothesis from section 8.1 and is not yet validated. The last column states how each difference would be measured in a usability study.

Stage	Today (hypothesis)	With Adeed (prototype)	How to measure it
Understanding the report	The patient reads technical metrics alone and may ask someone else for help.	Metrics are explained in plain language with their source and data-quality notes.	Comprehension questions on a de-identified sample report.
Finding past results	Reports and lab results are searched for manually.	Reports and HbA1c results are kept in one history.	Time to find the previous report.
Judging comparability	The patient may not know whether two reports can be compared.	The system checks comparability and explains limits.	Correct and incorrect comparisons made by users.

Stage	Today (hypothesis)	With Adeed (prototype)	How to measure it
Preparing questions	Questions are remembered or written ad hoc.	Suggested, editable questions with the reason for each.	Number and relevance of questions prepared.
Taking information to the visit	Printouts or memory.	One-page appointment summary.	User rating and clinician feedback.
Professional review	Depends on the visit.	Structured review request (simulated in the prototype).	Completeness of the information a reviewer receives.

9. Wireframes and Product Screens

The wireframes below show the intended end-to-end design: account selection, home, profile, report log, report details, HbA1c analysis, comparison, doctor questions, appointment preparation, appointment booking, the Sukkarah chat, and the PDF export. The working prototype in sections 6-8 implements part of this design.



Figure 12. *Twelve wireframes of the proposed Adeed experience.*

10. Expected Business and User Impact

10.1 Expected User Benefits

If the underlying need is validated and the system performs reliably, potential benefits include easier access to report information, more understandable historical comparisons, reduced effort in

organizing results, better-structured appointment preparation, and clearer visibility into review status. These are expected benefits, not proven outcomes.

10.2 Expected Organizational Benefits

Potential organizational benefits include more structured information submitted for review, better visibility into pending requests, and a more organized follow-up workflow. Real impact would depend on adoption, clinical review capacity, data quality, integration quality, and governance.

10.3 Success Metrics

Numerical targets should be agreed before evaluation and justified by the project scope. Do not report a target as an achieved result. Any claims about clinical effectiveness require appropriate clinical validation and are outside what a basic course prototype can establish. Candidate measures, with no targets set yet:

- Task completion and time to complete a report-interpretation task.
- Comprehension score on questions about a sample report.
- Number and relevance of appointment questions prepared.
- Share of extracted values flagged as uncertain, and how often users correct them.
- User-rated clarity and trust (for example, a short post-task survey).
- Completeness of the information a reviewer receives with a request.

11. Limitations and Risk Management

The prototype should not be presented as a clinically validated product. Any real-world deployment would require further security, privacy, clinical governance, regulatory, and integration review.

- **Data and privacy:** demonstration data only; real health data would need appropriate approvals and hosting decisions.
- **AI safety:** the assistant is bounded, refuses diagnosis and dosing, and shows an emergency notice; its answers still need evaluation for accuracy.
- **Evidence:** the user problem is a hypothesis; social-media and forum posts are anecdotal.
- **Review workflow:** simulated in the prototype and not connected to any real clinician.
- **Scope:** supported report formats are limited, and unsupported files may fail or produce uncertain values.

12. Conclusion

Adeed proposes a focused AI-assisted workflow for organizing glucose reports, comparing current measurements with compatible historical results, explaining verified information, and supporting professional follow-up. The concept responds to a relevant public-health context while recognizing that the specific user problem still needs direct validation.

Its central design principle is to combine deterministic data processing with constrained AI assistance and human clinical oversight. By making sources, units, missing data, and comparison limitations visible, the proposed service aims to support clearer patient understanding without

replacing medical judgment. The next steps are to validate the problem with intended users, test the report-processing workflow, evaluate usability and safety, and verify any integration assumptions.

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